



Autocorrelation: What Happens If the Error Terms Are Correlated?

Damodar N. Gujarati and Dawn C Porter (2009)

- **Learning Objectives:**

- 1. What is the nature of autocorrelation?
- 2. What are the theoretical and practical consequences of autocorrelation?
- 3. How does one know that there is autocorrelation in any given situation?
- 4. How does one remedy the problem of autocorrelation?

12.1 The Nature of the Problem

- **Contemporaneous Correlation:** Linear relationship between two variables in the same time period e.g., correlation between X_t and Y_t , current income and current consumption (C_t , Y_t).
- **Serial Correlation:** Correlation between the values of one variable in one time period and another or same variable in a different time period e.g., C_t and C_{t-1} , C_t and Y_{t-1} , or K_t and K_{t-1} .
- **Autocorrelation:** A special form of serial correlation where in current value of a variable are correlated with lagged value of same variable e.g., C_t and C_{t-1} .

12.1 The Nature of the Problem Conti....

- There are different types of serial correlation. With *first-order serial correlation*, errors in one time period are correlated directly with errors in the ensuing time period.
- With *positive serial correlation*, errors in one time period are positively correlated with errors in the next time period.

Definition: First-order of Autocorrelation, AR(1)

$$Y_t = \beta_1 + \beta_2 X_{2t} + u_t \quad t = 1, \dots, T$$

If $\text{Cov}(u_t, u_s) = E(u_t u_s) \neq 0$ where $t \neq s$

and if

$$u_t = \rho u_{t-1} + v_t$$

where $-1 < \rho < 1$ (ρ : RHO)

and $v_t \sim \text{iid}(0, \sigma_v^2)$ (white noise)

$$E(v_t) = 0$$

$$\text{var}(v_t) = \sigma_v^2$$

$$\text{cov}(v_t, v_s) = 0$$

This scheme is called **first-order** autocorrelation and denotes as **AR(1)**

Autoregressive : The regression of u_t can be explained by itself lagged one period.

ρ (RIIO) : the first-order autocorrelation coefficient
or 'coefficient of autocovariance'

Example of serial correlation:

Year	$\text{Consumption}_t = \beta_1 + \beta_2 \text{Income}_t + \text{error}_t$		
1973	230	320	u_{1973}
...	...	...	
...	...	...	
1995	558	714	u_{1995}
1996	699	822	u_{1996}
1997	881	907	u_{1997}
1998	925	1003	u_{1998}
1999	984	1174	u_{1999}
2000	1072	1246	u_{2000}

Error term represents other factors that affect consumption

TaxRate₁₉₉₉

TaxRate₂₀₀₀

The current year Tax Rate may be determined by previous year rate

$$\text{TaxRate}_{2000} = \rho \text{TaxRate}_{1999} + v_{2000}$$

$$\Rightarrow \boxed{u_t = \rho u_{t-1} + v_t} \quad \leftarrow v_t \sim \text{iid}(0, \sigma_v^2)$$

Why does serial correlation occur?

Inertia. A salient feature of most economic time series is inertia, or sluggishness. As is well known, time series such as GNP, price indexes, production, employment, and unemployment exhibit (business) cycles.

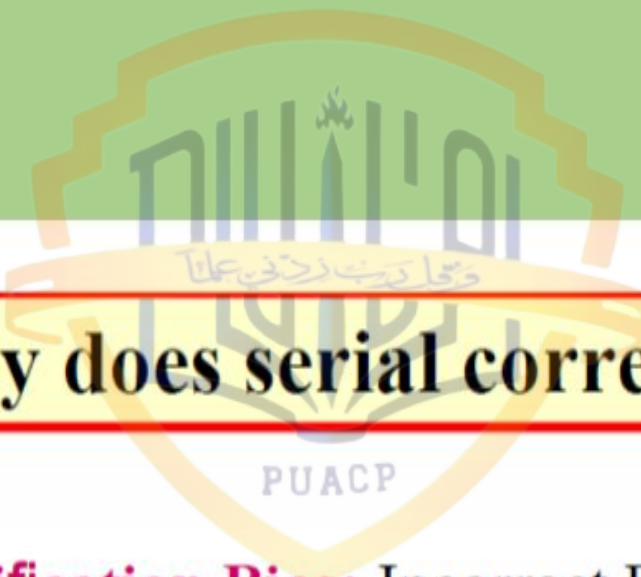
Specification Bias: Excluded Variables Case. This is the case of excluded variable specification bias. Often the inclusion of such variables removes the correlation pattern observed among the residuals. For example, suppose we have the following demand model:

$$Y_t = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + \beta_4 X_{4t} + u_t \quad (12.1.2)$$

where Y = quantity of beef demanded, X_2 = price of beef, X_3 = consumer income, X_4 = price of pork, and t = time.⁴ However, for some reason we run the following regression:

$$Y_t = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + v_t \quad (12.1.3)$$

Now if Eq. (12.1.2) is the “correct” model or the “truth” or true relation, running Eq. (12.1.3) is equivalent to letting $v_t = \beta_4 X_{4t} + u_t$. And to the extent the price of pork affects the consumption of beef, the error or disturbance term v will reflect a systematic pattern, thus creating (false) autocorrelation.

The logo of the Palestinian University of Applied Sciences (PUACP) is centered in the background. It features a shield with Arabic calligraphy and the acronym 'PUACP' below it.

Why does serial correlation occur?

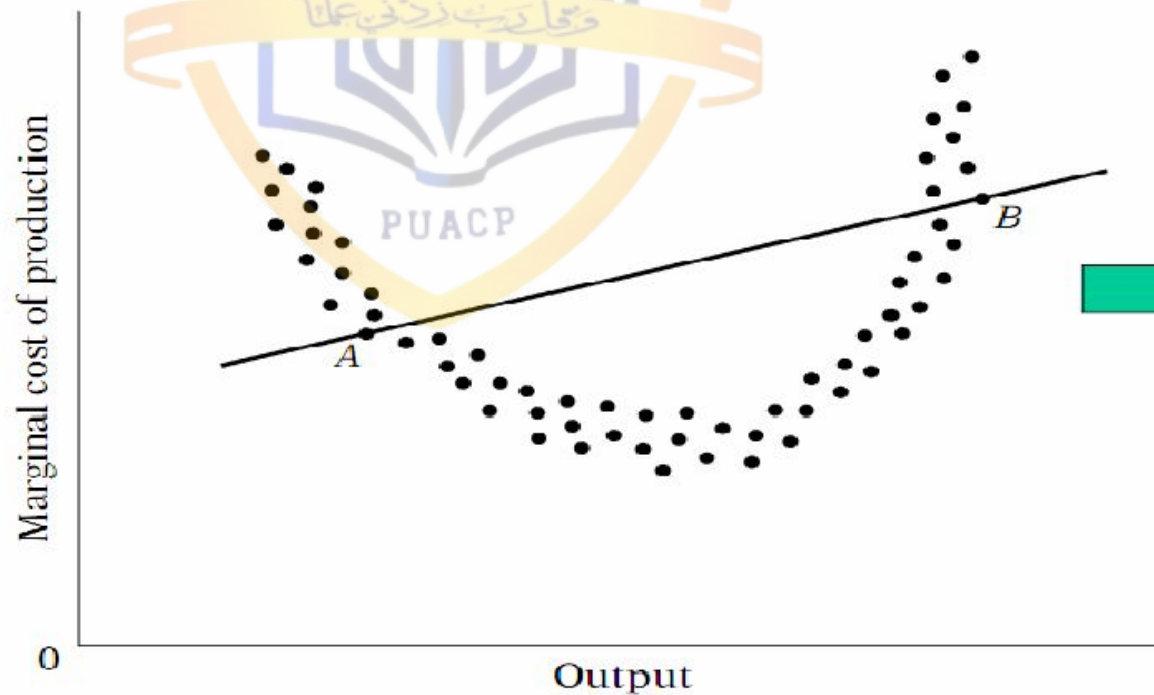
Specification Bias: Incorrect Functional Form. Suppose the “true” or correct model in a cost-output study is as follows:

$$\text{Marginal cost}_i = \beta_1 + \beta_2 \text{ output}_i + \beta_3 \text{ output}_i^2 + u_i$$

but we fit the following model:

$$\text{Marginal cost}_i = \alpha_1 + \alpha_2 \text{ output}_i + v_i$$

Why does serial correlation occur?



Specification bias: incorrect functional form.

Why does serial correlation occur?

Cobweb Phenomenon. The supply of many agricultural commodities reflects the so-called cobweb phenomenon, where supply reacts to price with a lag of one time period because supply decisions take time to implement (the gestation period).

$$\text{Supply}_t = \beta_1 + \beta_2 P_{t-1} + u_t$$

Lags. In a time series regression of consumption expenditure on income, it is not uncommon to find that the consumption expenditure in the current period depends, among other things, on the consumption expenditure of the previous period. That is,

$$\text{Consumption}_t = \beta_1 + \beta_2 \text{income}_t + \beta_3 \text{consumption}_{t-1} + u_t$$

Why does serial correlation occur?

“Manipulation” of Data. In empirical analysis, the raw data are often “manipulated.”

Example: Converting monthly data to quarterly introduces smoothness into the data by dampening the fluctuations in the monthly data.

Therefore, the graph plotting the quarterly data looks much smoother than the monthly data, and this smoothness may itself lend to a systematic pattern in the disturbances, thereby introducing autocorrelation.

Data Transformation.

$$Y_t = \beta_1 + \beta_2 X_t + u_t \quad \longrightarrow \quad \Delta Y_t = \beta_2 \Delta X_t + v_t$$
$$v_t = \Delta u_t = (u_t - u_{t-1})$$

it can be shown that the error term v_t is autocorrelated.

Why does serial correlation occur?

Nonstationarity. a time series is stationary if its characteristics (e.g., mean, variance, and covariance) are time invariant; that is, they do not change over time. If that is not the case, we have a nonstationary time series.

If $u_t = \rho_1 u_{t-1} + v_t$
it is AR(1), **first-order** autoregressive

If $u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + v_t$
it is AR(2), **second-order** autoregressive

.....
If $u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \dots + \rho_p u_{t-p} + v_t$
it is AR(p), **pth-order** autoregressive

High order
autocorrelation

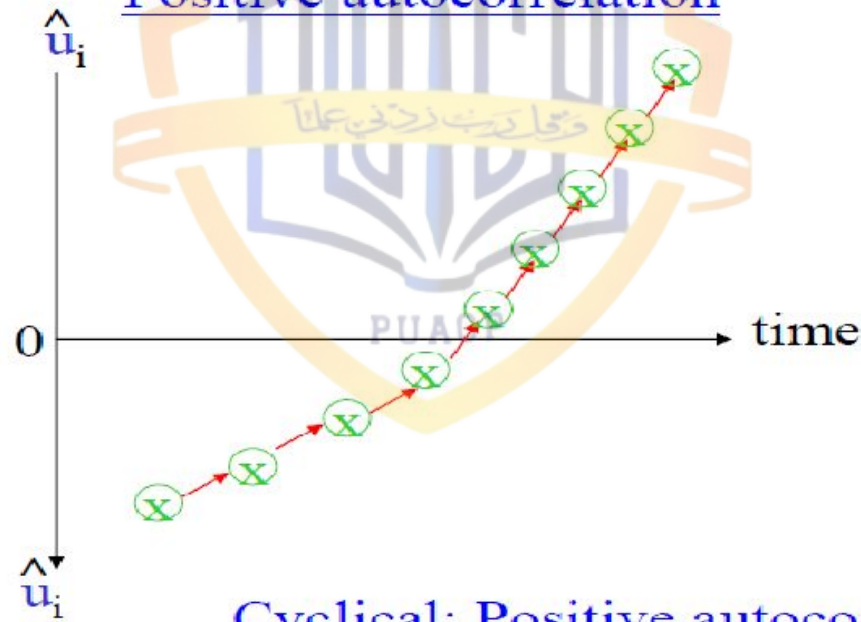
Autocorrelation AR(1) :

$\text{Cov}(u_t, u_{t-1}) > 0 \Rightarrow 0 < \rho < 1$ **positive** AR(1)

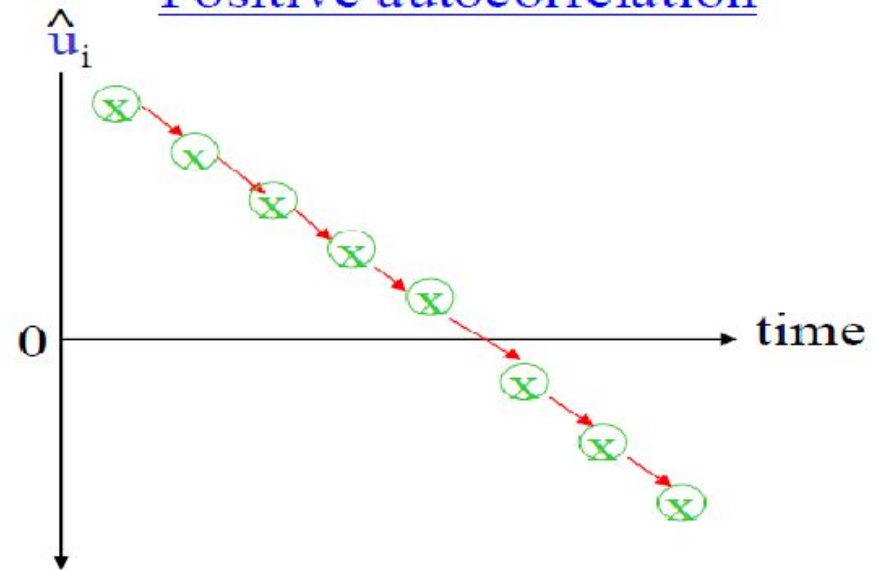
$\text{Cov}(u_t, u_{t-1}) < 0 \Rightarrow -1 < \rho < 0$ **negative** AR(1)

$$-1 < \rho < 1$$

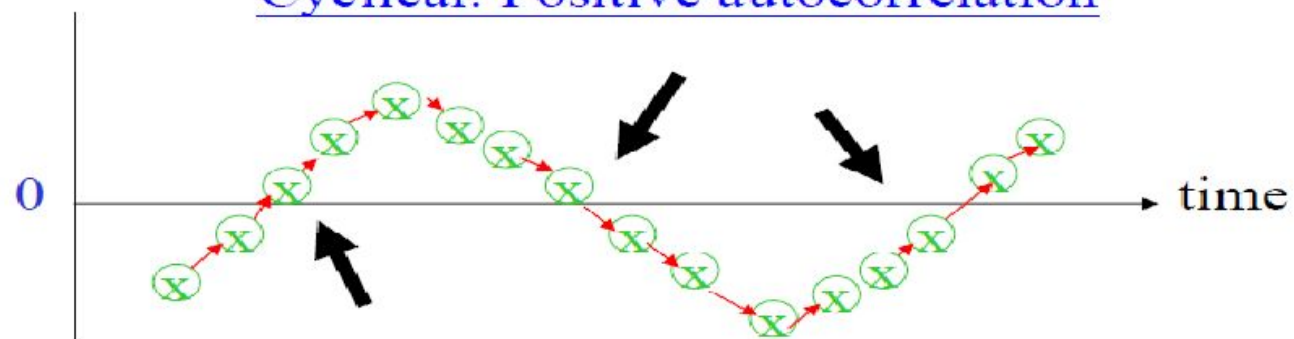
Positive autocorrelation



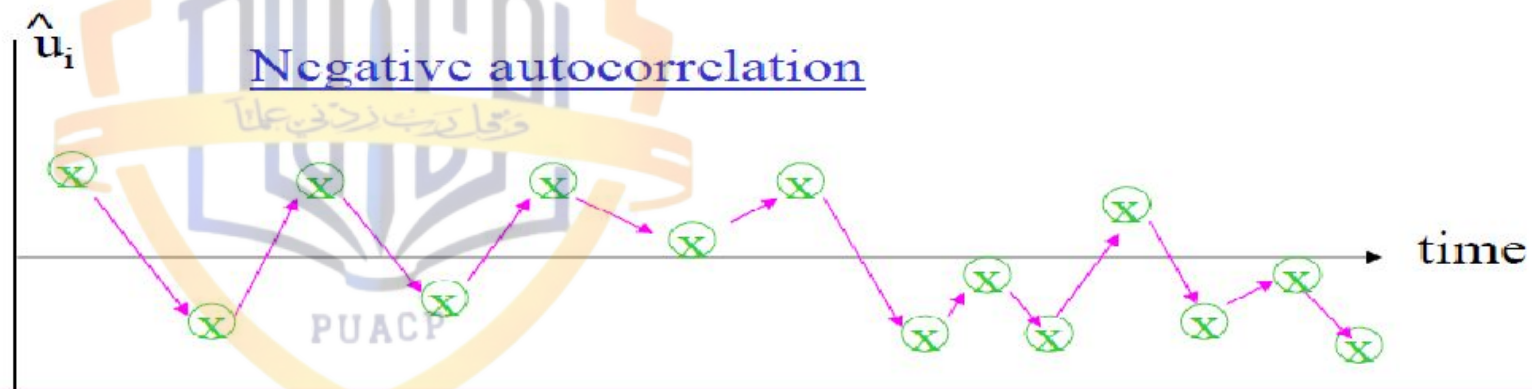
Positive autocorrelation



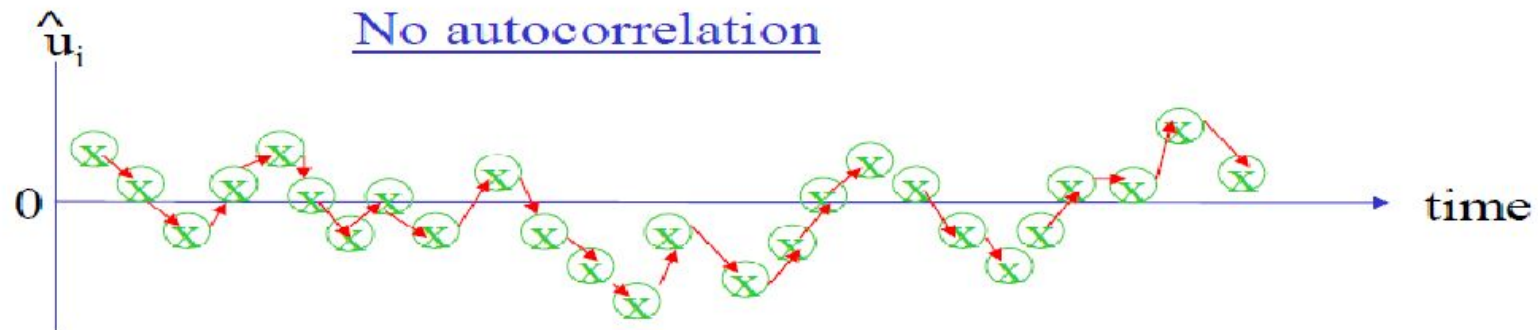
Cyclical: Positive autocorrelation



The current error term tends to have the same sign as the previous one.



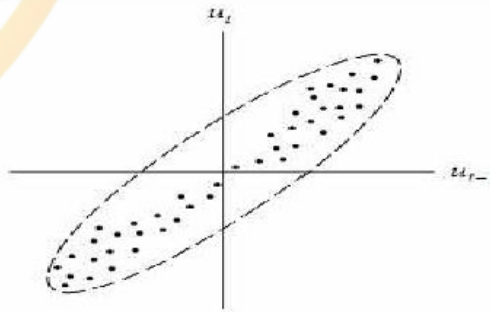
The current error term tends to have the opposite sign from the previous.



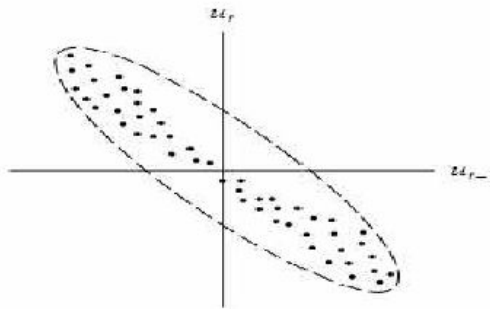
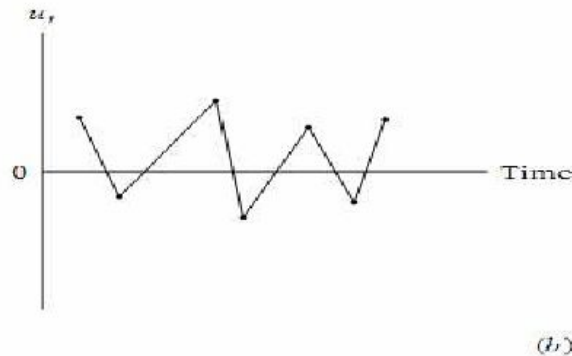
The current er. term tends to be randomly moving wrt to the previous term

12.2 OLS Estimation in the Presence of Autocorrelation

Positive vs. Negative Autocorrelation



$$u_t = \rho u_{t-1} + \varepsilon_t$$
$$\rho > 0$$



$$u_t = \rho u_{t-1} + \varepsilon_t$$
$$\rho < 0$$

(a) Positive and (b) negative autocorrelation.

The meaning of ρ : The error term u_t at time t is a linear combination of the current and past disturbance.

$$0 < \rho < 1$$

$$-1 < \rho < 0$$

$$\rho = 1$$

$$\rho > 1$$

The further the period is in the past, the smaller is the weight of that error term (u_{t-1}) in determining u_t , past is less important than current

The past is as equally important as the current.

The past is more importance than the current.

The consequences of serial correlation:

1. The estimated coefficients are still unbiased.

$$E(\hat{\beta}_k) = \beta_k$$

2. The variances of the $\hat{\beta}_k$ is no longer the smallest

3. The standard error of the estimated coefficient, $se(\hat{\beta}_k)$ becomes large

**Therefore, when the regression has AR(1) errors,
The estimators are not BLUE.**

Two variable regression model: $Y_t = \beta_1 + \beta_2 X_{2t} + u_t$

The OLS estimator of β_2 , $\implies \hat{\beta}_2 = \frac{\sum \mathbf{x} \mathbf{y}}{\sum \mathbf{x}^2}$

$$\text{If } E(u_t u_{t-1}) = 0 \quad \text{then} \quad \text{Var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum \mathbf{x}_t^2}$$

If $E(u_t u_{t-1}) \neq 0$, and $u_t = \rho u_{t-1} + v_t$, then

$$\text{var}(\hat{\beta}_2)_{\text{AR1}} = \frac{\sigma^2}{\sum \mathbf{x}_t^2} + \frac{2\sigma^2}{\sum \mathbf{x}_t^2} \left[\rho \frac{\sum \mathbf{x}_t \mathbf{x}_{t+1}}{\sum \mathbf{x}_t^2} + \rho^2 \frac{\sum \mathbf{x}_t \mathbf{x}_{t+2}}{\sum \mathbf{x}_t^2} + \dots \right]$$

$$-1 < \rho < 1$$

If $\rho = 0$, zero autocorrelation, then $\text{Var}(\hat{\beta}_2)_{\text{AR1}} = \text{Var}(\hat{\beta}_2)$

If $\rho \neq 0$, autocorrelation, then $\text{Var}(\hat{\beta}_2)_{\text{AR1}} \neq \text{Var}(\hat{\beta}_2)$

The AR(1) variance is not correct
and may not be the smallest,

Autoregressive scheme:

$$u_t = \rho u_{t-1} + v_t$$

$$\Rightarrow u_{t-1} = \rho u_{t-2} + v_{t-1}$$

$$\Rightarrow u_{t-2} = \rho u_{t-3} + v_{t-2}$$

$$\Rightarrow u_t = \rho[\rho u_{t-2} + v_{t-1}] + v_t$$

$$u_t = \rho^2 u_{t-2} + \rho v_{t-1} + v_t$$

$$\Rightarrow u_t = \rho^2 [\rho u_{t-3} + v_{t-2}] + \rho v_{t-1} + v_t$$

$$u_t = \rho^3 u_{t-3} + \rho^2 v_{t-2} + \rho v_{t-1} + v_t$$

$$\sigma^2 = \text{var}(u_t) = E(u_t u_t) = \frac{\sigma_v^2}{1 - \rho^2}$$

$$E(u_t u_{t-1}) = \rho \sigma^2$$

$$E(u_t u_{t-2}) = \rho^2 \sigma^2$$

.....

$$E(u_t u_{t-s}) = \rho^s \sigma^2$$

It means the more we go to past,
the less effect on current period

Gujarati(2003) Table12.4, pp.460

How to detect autocorrelation ?

How to detect autocorrelation ?

Equation: EQ01		Workfile:							
View	Procs	Objects	Print	Name	Freeze	Estimate	Forecast	Stats	Resids
Dependent Variable: Y									
Method: Least Squares									
Date: 02/17/03 Time: 16:40									
Sample: 1959 1998									
Included observations: 40									
Variable		Coefficient	Std. Error	t-Statistic	Prob.				
C		29.51925	1.942347	15.19773	0.0000				
X		0.713659	0.024105	29.60658	0.0000				
R-squared		0.958449	Mean dependent var		85.64500				
Adjusted R-squared		0.957356	S.D. dependent var		12.95632				
S.E. of regression		2.675533	Akaike info criterion		4.854881				
Sum squared resid		272.0220	Schwarz criterion		4.939325				
Log likelihood		-95.09761	F-statistic		876.5495				
Durbin-Watson stat		0.122904	Prob(F-statistic)		0.000000				



Run OLS: $\hat{u}_t = \rho \hat{u}_{t-1} + v_t$ and check the t-value of the coefficient

View	Procs	Objects	Print	Name	Freeze	Estimate	Forecast	Stats	Resids
Dependent Variable: RESID_1									
Method: Least Squares									
Date: 05/03/04 Time: 17:49									
Sample(adjusted): 1960 1998									
Included observations: 39 after adjusting endpoints									
Variable		Coefficient	Std. Error	t-Statistic	Prob.				
RESID_1(-1)		0.914245	0.056337	16.22811	0.0000				
R-squared		0.873615	Mean dependent var		0.120615				
Adjusted R-squared		0.873615	S.D. dependent var		2.561492				
S.E. of regression		0.910629	Akaike info criterion		2.675945				
Sum squared resid		31.51134	Schwarz criterion		2.718600				
Log likelihood		-51.18093	Durbin-Watson stat		1.472987				

$$0.914245 = \hat{\rho} \approx 1 - \frac{DW}{2} = 1 - \frac{0.122904}{2} = 0.9385$$

Durbin-Watson Autocorrelation test

From OLS regression result: where d or $DW^* = 0.1229$

Check DW Statistic Table (At 5% level of significance, $k' = k-1=1$, $n=40$)

$$d_L = 1.246$$

$$d_u = 1.344$$

Reject
 H_0
region

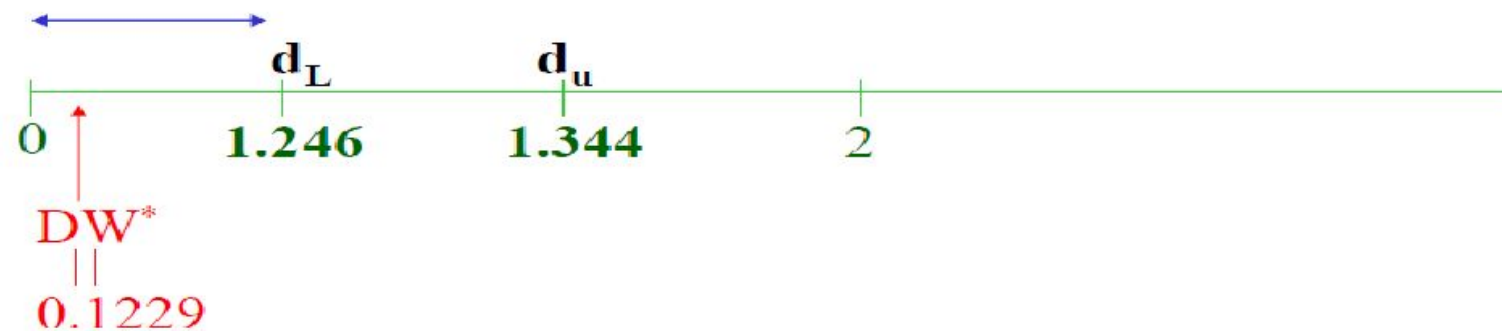
H_0 : no autocorrelation

$$\rho = 0$$

H_1 : yes, autocorrelation exists.

$$\text{or } \rho > 0$$

positive autocorrelation



Durbin-Watson test

OLS : $Y = \beta_1 + \beta_2 X_2 + \dots + \beta_k X_k + u_t$
obtain $\hat{u}_t$, DW-statistic(d)

Assuming AR(1) process: $u_t = \rho u_{t-1} + v_t$

$-1 < \rho < 1$	I. $H_0 : \rho = 0$	no autocorrelation
	$H_1 : \rho > 0$	yes, positive autocorrelation

DW^*

Compare d^* and d_L , d_u (critical values)

if $d^* < d_L \implies$ reject H_0

if $d^* > d_u \implies$ not reject H_0

if $d_L \leq d^* \leq d_u \implies$ this test is inconclusive

Durbin-Watson test(Cont.)

$$\text{DW} = \frac{\sum_{t=2}^T (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^T \hat{u}_t^2} \approx 2(1 - \hat{\rho})$$

(d)

$$\mathbf{d} = 2(1 - \hat{\rho})$$
$$\implies \frac{\mathbf{d}}{2} = 1 - \hat{\rho}$$

$$\implies \hat{\rho} = 1 - \frac{\mathbf{d}}{2}$$

Since $-1 \leq \hat{\rho} \leq 1$
implies $0 \leq \mathbf{d} \leq 4$

Durbin-Watson test(Cont.)

$$\text{II. } H_0 : \rho = 0$$

no negative autocorrelation

$$H_1 : \rho < 0$$

yes, negative autocorrelation

we use $(4 - d)$

(when d is greater than 2)

$$\text{if } (4 - d) < d_L$$

$$\text{or } 4 - d_L < d \implies \text{reject } H_0$$

$$\text{if } (4 - d) > d_u$$

$$\text{or } 4 - d_u > d \implies \text{not reject } H_0$$

$$\text{if } d_L \leq (4 - d) \leq d_u$$

$$\text{or } 4 - d_u \leq d \leq 4 - d_L \implies \text{inconclusive}$$

Durbin-Watson test(Cont.)

II. $H_0 : \rho = 0$

No autocorrelation

$H_1 : \rho \neq 0$

two-tailed test for auto correlation
either positive or negative AR(1)

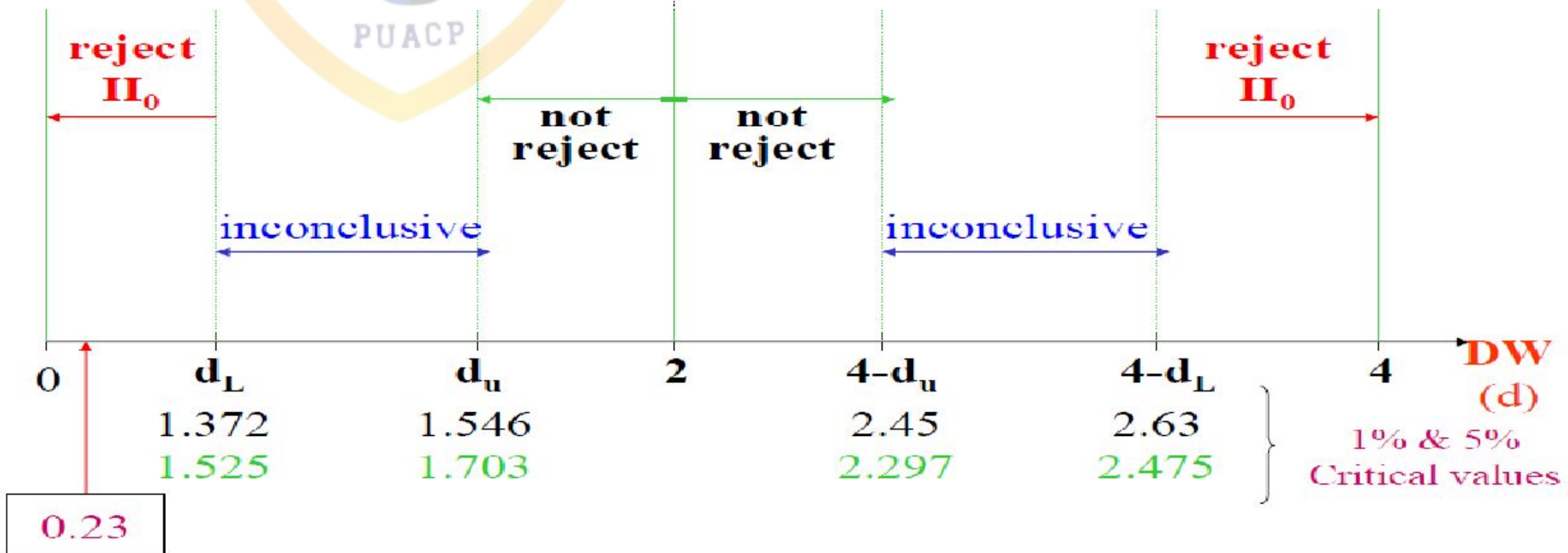
If $d < d_L$
or $d > 4 - d_L$ } $\implies$ reject H_0

If $d_u < d < 4 - d_u \implies$ not reject H_0

If $d_L \leq d \leq d_u$
or $4 - d_u \leq d \leq 4 - d_L$ } $\implies$ inconclusive

$H_0 : \rho = 0$
positive autocorrelation
 $H_1 : \rho > 0$

$H_0 : \rho = 0$
negative autocorrelation
 $H_1 : \rho < 0$



for example :

$$\hat{UM}_t = 23.1 - 0.078 CA_t$$

(15.6) (2.0)

$$\bar{R}^2 = 0.78 \quad F = 78.9 \quad \hat{\sigma}_u$$

$$\hat{UM}_t = 23.1 - 0.078 CAP_t - 0.146 CAP_{t-1} + 0.043 T_t$$

(15.6) (2.0) (3.7) (10.3)

$$\bar{R}^2 = 0.78 \quad F = 78.9 \quad \hat{\sigma}_u = 0.677 \quad \text{RSS} = 29.3 \quad \text{DW} = 0.23 \quad n = 68$$

- (i) $K' = 3$ (number of independent variable)
- (ii) $n = 68$, $\alpha = 0.01$ significance level
 0.05
- (iii) $d_L = 1.525$, $d_u = 1.703$ $\leftarrow 0.05$
 $d_L = 1.372$, $d_u = 1.546$ $\leftarrow 0.01$

Reject H_0 , positive autocorrelation exists

The assumptions underlying the d(DW) statistics :

1. Intercept term must be included in OLS regression.
2. **X's are nonstochastic**
3. Only test **AR(1)** : $u_t = \rho u_{t-1} + v_t$ where $v_t \sim \text{iid}(0, \sigma_v^2)$
4. **Not include the lagged dependent variable,**

$$Y_t = \beta_1 + \beta_2 X_t^2 + \beta_3 X_t^3 + \dots + \beta_k X_t^k + \gamma Y_{t-1} + u_t$$

(autoregressive model)

5. **No missing observation**

missing →

	Y	X
1970	100	15
⋮	⋮	⋮
1980	235	20
81	N.A.	N.A.
82	N.A.	N.A.
93	253	37
94	281	41
95	⋮	⋮

Breusch-Godfrey (BG) test of higher-order autocorrelation or called Durbin's m test (Lagrange Multiplier, LM, Test)

Test Procedures:

- (1) Run OLS and obtain the residuals $\hat{u}_t$.
- (2) Run $\hat{u}_t$ against all the regressors in the model plus the additional regressors, $\hat{u}_{t-1}, \hat{u}_{t-2}, \hat{u}_{t-3}, \dots, \hat{u}_{t-p}$.
$$\hat{u}_t = \beta_1 + \beta_2 X_t + \hat{u}_{t-1} + \hat{u}_{t-2} + \hat{u}_{t-3} + \dots + \hat{u}_{t-p} + v$$

Obtain the R^2 value from this regression.
- (3) compute the BG-statistic: $(n-p)R^2$
- (4) compare the BG-statistic to the χ^2_p (p is # of degree-order)
- (5) If $BG > \chi^2_p$, reject H_0 (No autocorrelation),
it means there is a higher-order autocorrelation
If $BG < \chi^2_p$, not reject H_0 ,
it means there is a no higher-order autocorrelation

Click
VIEW

Equation: EQ01 Workfile: TABLE12-4

Representations ne Freeze Estimate Forecast Stats Resids

Estimation Output

Actual, Fitted, Residual

Gradients and Derivatives

Covariance Matrix

Coefficient Tests

Residual Tests

Stability Tests

Label

R-squared

Adjusted R-squared

S.E. of regression

Sum squared resid

Log likelihood

Durbin-Watson stat

Correlogram - Q-statistics

Correlogram Squared Residuals

Histogram - Normality Test

Serial Correlation LM Test...

ARCH LM Test...

White Heteroskedasticity (no cross terms)

White Heteroskedasticity (cross terms)

Schwarz criterion

F-statistic

Prob(F-statistic)

Prob.

.0000

.0000

.64500

.95632

.54881

272.0220

-95.09761

0.122904

4.939325

876.5495

0.000000

Equation: EQ01 Workfile: TABLE12-4

View | Procs | Objects | Print | Name | Freeze | Estimate | Forecast | Stats | Resids

Breusch-Godfrey Serial Correlation LM Test:

F-statistic	35.63806	Probability	0.000000
Obs*R-squared	32.29690	Probability	0.000002

Test Equation:
 Dependent Variable: RESID
 Method: Least Squares
 Date: 02/17/03 Time: 18:28
 Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.341927	0.959333	0.356422	0.7237
X	-0.005213	0.012172	-0.428292	0.6711
RESID(-1)	1.002383	0.176774	5.670407	0.0000
RESID(-2)	-0.088800	0.249578	-0.355803	0.7242
RESID(-3)	-0.068404	0.252938	-0.270438	0.7885
RESID(-4)	0.065685	0.192698	0.340869	0.7353

R-squared	0.807422	Mean dependent var	-5.76E-15
Adjusted R-squared	0.779102	S.D. dependent var	2.641008
S.E. of regression	1.241268	Akaike info criterion	3.407624
Sum squared resid	52.38534	Schwarz criterion	3.660956
Log likelihood	-62.15249	F-statistic	28.51044
Durbin-Watson stat	1.514323	Prob(F-statistic)	0.000000

Compare with
the critical values

Check on the
t-statistics
To see
The order of
autocorrelation

Remedy: 1. First-difference transformation

$$Y_t = \beta_1 + \beta_2 X_t + u_t$$

$$Y_{t-1} = \beta_1 + \beta_2 X_{t-1} + u_{t-1}$$

assume $\rho = 1$

$$\Rightarrow Y_t - Y_{t-1} = \beta_1 - \beta_1 + \beta_2 (X_t - X_{t-1}) + (u_t - u_{t-1})$$

$$\Rightarrow \Delta Y_t = \beta_2 \Delta X_t + \Delta u_t$$

no intercept

$$Y_t = \beta_1 + \beta_2 X_t + \beta_3 T + u_t$$

2. Add $T = \text{trend}$

$$Y_{t-1} = \beta_1 + \beta_2 X_{t-1} + \beta_3 (T-1) + \varepsilon_{t-1}$$

$$\Rightarrow (Y_t - Y_{t-1}) = (\beta_1 - \beta_1) + \beta_2 (X_t - X_{t-1}) + \beta_3 [T - (T-1)] + (u_t - u_{t-1})$$

$$\Rightarrow \Delta Y_t = \beta_2 \Delta X_t + \beta_3 * 1 + u'_t$$

$$\Rightarrow \Delta Y_t = \beta_1^* + \beta_2 \Delta X_t + u'_t$$

If $\hat{\beta}_1^* > 0 \Rightarrow$ an upward trend in Y

$$(\hat{\beta}_2^* > 0)$$

3. Cochrane-Orcutt Two-step procedure (CORC)

- (1). Run OLS on
and obtains $\hat{u}_t$

$$Y_t = \beta_1 + \beta_2 X_t + u_t$$

- (2). Run OLS on
and obtains $\hat{\rho}$

$$\hat{u}_t = \rho \hat{u}_{t-1} + v_t$$

- (3). Use the $\hat{\rho}$ to transform the variables :

$$Y_t^* = Y_t - \hat{\rho} Y_{t-1}$$

$$X_t^* = X_t - \hat{\rho} X_{t-1}$$

$$\begin{aligned} Y_t &= \beta_1 + \beta_2 X_t + u_t \\ \rightarrow \hat{\rho} Y_{t-1} &= \beta_1 \hat{\rho} + \beta_2 \hat{\rho} X_{t-1} + \hat{\rho} u_{t-1} \\ \hline (Y_t - \hat{\rho} Y_{t-1}) &= \beta_1(1 - \hat{\rho}) + \beta_2(X_t - \hat{\rho} X_{t-1}) + (u_t - \hat{\rho} u_{t-1}) \end{aligned}$$

- (4). Run OLS on

$$Y_t^* = \beta_1^* + \beta_2^* X_t^* + u_t^*$$

Generalized
Least Squares
(GLS)
method



4. Cochrane-Orcutt Iterative Procedure

(5). If DW test shows that the autocorrelation still existing, than it needs to iterate the procedures from (4). Obtain the

$$u_t^* = Y_t - \beta_1^* - \beta_2^* X_t$$

(6). Run OLS

$$\hat{u}_t^* = \hat{\rho} \hat{u}_{t-1}^* + v_t$$

$$\hat{\rho} \approx (1 - \frac{DW_2}{2})$$

and obtain $\hat{\rho}$ which is the second-round estimated ρ

(7). Use the $\hat{\rho}$ to transform the variable

$$Y_t^{**} = Y_t - \hat{\rho} Y_{t-1}$$

$$X_t^{**} = X_t - \hat{\rho} X_{t-1}$$

$$Y_t = \beta_1 + \beta_2 X_t + u_t$$

$$\hat{\rho} Y_{t-1} - \beta_1 \hat{\rho} + \beta_2 \hat{\rho} X_{t-1} + \hat{\rho} u_{t-1}$$

(8). Run OLS on

$$Y_t^{**} = \beta_1^{**} + \beta_2^{**} X_t^{**} + u_t^{**}$$

Where is $(Y_t - \hat{\rho} Y_{t-1}) = \beta_1 (1 - \hat{\rho}) + \beta_2 (X_t - \hat{\rho} X_{t-1}) + (u_t - \hat{\rho} u_{t-1})$

Cochrane-Orcutt Iterative procedure(Cont.)

(9). Check on the DW_3 -statistic, if the autocorrelation is still exist, than go into the third-round and so on.

Until $(\hat{\hat{\rho}} - \hat{\hat{\rho}} < 0.01)$, the estimated ρ 's differ a little.

Generalized least Squares (GLS)

5. Prais-Winsten transformation

$$Y_t = \beta_1 + \beta_2 X_t + u_t \quad t = 1, \dots, T \quad (1)$$

Assume AR(1) : $u_t = \rho u_{t-1} + v_t \quad -1 < \rho < 1$

$$\rho Y_{t-1} = \rho \beta_1 + \rho \beta_2 X_{t-1} + \rho u_{t-1} \quad (2)$$

$$(1) - (2) \Rightarrow (Y_t - \rho Y_{t-1}) = \beta_1 (1 - \rho) + \beta_2 (X_t - \rho X_{t-1}) + (u_t - \rho u_{t-1})$$

$$\text{GLS} \Rightarrow \underline{Y_t^*} = \underline{\beta_1^*} + \underline{\beta_2^*} \underline{X_t^*} + u_t^*$$

To avoid the loss of the first observation of each variable, **the first observation** of Y^* and X^* should be transformed as :

$$Y_{t=1}^* = \sqrt{1 - \hat{\rho}^2} (Y_{t=1})$$

$$X_{t=1}^* = \sqrt{1 - \hat{\rho}^2} (X_{t=1})$$

but $Y_{t=2}^* = Y_{t=2} - \hat{\rho} Y_{t=1}$; $X_{t=2}^* = X_{t=2} - \hat{\rho} X_{t=1}$

$$Y_{t-3}^* = Y_{t-3} - \hat{\rho} Y_{t-2} ; X_{t-3}^* = X_{t-3} - \hat{\rho} X_{t-2}$$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

$$Y_t^* = Y_t - \hat{\rho} Y_{t-1} ; X_t^* = X_t - \hat{\rho} X_{t-1}$$

6. Durbin's Two-step method :

$$Y_t = \beta_1 + \beta_2 X_t + u_t$$

Since $(Y_t - \rho Y_{t-1}) = \beta_1 (1 - \rho) + \beta_2 (X_t - \rho X_{t-1}) + V_t$

$$\Rightarrow Y_t = \beta_1^* + \beta_2 X_t - \rho \beta_2 X_{t-1} + \rho Y_{t-1} + V_t$$

I. Run OLS
this specification

$$Y_t - \beta_1^* + \beta_2^* X_t - \beta_3^* X_{t-1} + \beta_4^* Y_{t-1} + v_t$$

Obtain $\hat{\beta}_4^*$ as an estimated $\hat{\rho}$ (RHO)

II. Transforming the variables :

$$Y_t^* = Y_t - \hat{\beta}_4^* Y_{t-1} \quad \text{as} \quad Y_t^* = Y_t - \hat{\rho} Y_{t-1}$$

and $X_t^* = X_t - \hat{\beta}_4^* X_{t-1} \quad \text{as} \quad X_t^* = X_t - \hat{\rho} X_{t-1}$

III. Run OLS on model :

$$Y_t^* = \alpha_1 + \alpha_2 X_t^* + u'_t$$

where $\hat{\alpha}_1 = \hat{\beta}_1(1 - \rho)$ and $\hat{\alpha}_2 = \hat{\beta}_2$

Compare

Lagged Dependent Variable and Autocorrelation

$$Y_t = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + \dots + \beta_k X_{k,t} + \alpha_1 Y_{t-1} + u_t$$

DW statistic will often be close to 2 or

DW does not converge to $2(1 - \hat{\rho})$

DW is not reliable

Durbin-h Test:

Compute

$$h^* = \hat{\rho} \times \sqrt{\frac{n}{1 - n \cdot \text{var}(\hat{\alpha}_1)}}$$

Compare h^* to Z where $Z^c \sim N(0,1)$ normal distribution

If $|h^*| > Z^c \Rightarrow \text{reject } H_0 : \rho = 0 \text{ (no autocorrelation)}$

Example: Gujarati(2003) Table 12-4, p.460

Equation: UNTITLED Workfile: TABLE12-4				
View Procs Objects Print Name Freeze Estimate Forecast Stats Resids				
Dependent Variable: Y				
Method: Least Squares				
Date: 02/17/03 Time: 16:40				
Sample: 1959 1998				
Included observations: 40				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	29.51925	1.942347	15.19773	0.0000
X	0.713659	0.024105	29.60658	0.0000
R-squared	0.958449	Mean dependent var	85.64500	
Adjusted R-squared	0.957356	S.D. dependent var	12.95632	
S.E. of regression	2.675533	Akaike info criterion	4.854881	
Sum squared resid	272.0220	Schwarz criterion	4.939325	
Log likelihood	-95.09761	F-statistic	876.5495	
Durbin-Watson stat	0.122904	Prob(F-statistic)		

$$\text{Wage}(Y_t) = \beta_1 + \beta_2 \text{Output}(X_t) + u_t$$

$$\text{DW} \approx 2(1 - \hat{\rho})$$
$$0.1229 \approx 2(1 - 0.9385)$$

View Procs Objects Print Name Freeze Estimate Forecast

Dependent Variable: RESID_1
 Method: Least Squares
 Date: 05/03/04 Time: 17:49
 Sample(adjusted): 1960 1998
 Included observations: 39 after adjusting endpoints

$$u_t = \rho u_{t-1} + v_t$$

Variable	Coefficient	Std. Error	t-Statistic	Prob.
RESID_1(-1)	0.914245	0.056337	16.22811	0.0000
R-squared	0.873615	Mean dependent var	0.120615	
Adjusted R-squared	0.873615	S.D. dependent var	2.561492	
S.E. of regression	0.910629	Akaike info criterion	2.675945	
Sum squared resid	31.51134	Schwarz criterion	2.718600	
Log likelihood	-51.18093	Durbin-Watson stat	1.472987	

$$\hat{\rho} \approx 1 - \frac{DW}{2}$$

$$0.9142 \approx (1 - 0.0614)$$

Cochrane-Orcutt Two-step procedure (2)

Equation: EQ05 Workfile: TABLE12-4				
View	Procs	Objects	Print	Name
Dependent Variable: $Y-0.914245 \cdot Y(-1)$				
Method: Least Squares				
Date: 05/03/04 Time: 17:57				
Sample(adjusted): 1960 1998				
Included observations: 39 after adjusting endpoints				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	4.108209	0.656933	6.253621	0.0000
$X-0.914245 \cdot X(-1)$	0.528896	0.077424	6.831186	0.0000
R-squared	0.557761	Mean dependent var	8.496580	
Adjusted R-squared	0.545808	S.D. dependent var	1.273241	
S.E. of regression	0.858085	Akaike info criterion	2.581693	
Sum squared resid	27.24346	Schwarz criterion	2.667004	
Log likelihood	-48.34301	F-statistic	46.66510	
Durbin-Watson stat	1.620601	Prob(F-statistic)	0.000000	

Critical values:

$$D_u = 1.337$$

$$D_L = 1.237$$

Since $DW > D_u$

$\Rightarrow$ No

Autocorrelation
After CO-correction

Running the Cochrane-Orcutt iterative procedure in EVIEWS

Equation Specification

Equation specification
Dependent variable followed by list of regressors including ARMA and PDL terms, OR an explicit equation like $Y=c(1)+c(2)*X$.

y c x AR(1)

Estimation settings

Method: LS - Least Squares (NLS and ARMA)

Sample: 1959 1998

OK
Cancel
Options

Equation: EQ03 Workfile: TABLE12-4				
View Procs Objects Print Name Freeze Estimate Forecast Stats Resids				
Dependent Variable: Y				
Method: Least Squares				
Date: 02/17/03 Time: 17:42				
Sample(adjusted): 1960 1998				
Included observations: 39 after adjusting endpoints				
Convergence achieved after 7 iterations				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	45.03925	8.081898	5.572855	0.0000
X	0.550889	0.079019	6.971646	0.0000
AR(1)	0.891345	0.042583	20.93178	0.0000
R-squared	0.995339	Mean dependent var		86.34103
Adjusted R-squared	0.995080	S.D. dependent var		12.34486
S.E. of regression	0.865881	Akaike info criterion		2.623665
Sum squared resid	26.99100	Schwarz criterion		2.751631
Log likelihood	-48.16147	F-statistic		3843.976
Durbin-Watson stat	1.603995	Prob(F-statistic)		0.000000
Inverted AR Roots	.89			

$\hat{\rho}$